

RESPONSES SPECIFIC TO USE CASES DEFINED BELOW

The government would like to assess RFI responses based on specific use cases. This section details key use cases that exemplify NGA's operational requirements for an Automated Feature Extraction (AFE) solution. Responding vendors shall select one, two, or more features from the comprehensive list provided in Appendix A of the SOO. For each feature selected, the vendor must describe their solution's capabilities in the context of the most relevant use case category outlined below. The response should specifically detail the solution's proficiency across the three primary AFE functions: (A) Object Detection, (B) Geometry Extraction, and (C) Attribution, as defined in this section.

AFE RFI Features and Categories individual use cases:

1) Buildings

- a. Including agglomerations as Facility, Military Installation, or Built-Up Area
- b. Rationale: While the extraction of building geometries is a common capability, achieving consistent results with rich, functional attribution remains a significant challenge.
- c. Example Use Case: Identify all the built structures in a city and identify those that are places of worship.
- d. Use Case Objectives:
 - i. Global building footprints that accurately represent individual buildings at their ground-level footprint in vector format (e.g., splitting polygons along shared walls, polygons match the exterior walls).
 - ii. Characterization of Buildings into functional types aligning with TDS v7.1 and MGCP TRD functional categories.
 - iii. Methods to associate buildings with each other into facilities, installations, or built-up areas agglomerations.

2) Linear Transportation Features (Road, Railway, Cart Track, Trail)

- a. Rationale (internal): Long linear features present unique, but associated challenges for CV algorithms. The ontological differences between Road, Cart Track, and Trail are often unclear so approaching this problem as a group is likely more efficient than trying to overly parse by feature type.
- b. Example Use Case: Determine potential transportation routes between Point A and Point B.
- c. Use Case Objectives:
 - i. Extract a consistent, topologically correct transportation network that reflects the real-world transportation system as linear vector geometry, either along road centerlines or along ways of travel.
 - ii. Characterize physical attributes of the transportation features that influence navigability (e.g., width, surface composition, weather restriction).
 - iii. Characterize attributes about Transportation Feature use dynamics that are observable from behavior patterns (e.g., roadway type, one-way streets, lane counts, speed limits).

3) Discrete Transportation Features (Bridge, Ford, Tunnel, Gate)

- a. Rationale (internal): Transportation networks are typically most restricted by discrete features that limit what vehicles can navigate the network. These features all need to be associated with the network but are notionally independent of the transportation features and can use different detection methods.
 - b. Example Use Case: Determine if a vehicle (with specific height and weight) can navigate through choke points between Point A and Point B.
 - c. Use Case Objectives:
 - i. Identify and extract vector geometries (points, lines, and polygons) that represent real-world features that are associated with linear transportation ways.
 - ii. Characterize attributes of these features that influence successful transportation through these features (e.g., clearance height, load capacity, width)
 - iii. Associate these features with separately generated linear transportation geometries (e.g., have the road go through the tunnel or on the bridge).
- 4) Utilities (Cistern, Pylons, Cables, Electric Power Station, Pipeline, Power substation, Water Tower, Water Well)
- a. Rationale (internal): Utilities consist of both discrete and connected linear features but are all in association. In traditional manual collection many of these features can only be identified because of these associations (e.g., cables and pylons). For this reason, we can look at these as a collective problem-set.
 - b. Example Use Case: Determine what utilities connect to and serve a particular facility.
 - c. Use Case Objectives:
 - i. Identify and extract connected topologically correct utility networks (power, water, communications), with particular focus on longer distance services (e.g., high-voltage electrical cables rather than residential service lines).
 - ii. Characterize the attributes of extracted features, especially their association with certain types of utilities.
 - iii. Characterize the physical attributes (esp. height) of these features.
- 5) Built Water Infrastructure (Ditch, Culvert, Inland Waterbodies, Dam, Shoreline)
- a. Rationale (internal): Natural water features are collected through traditional LULC image classification efforts. But many of the constructed water features do not get picked up or get classified as simply "Water." Because hydrological features are all affected by elevation they can often be identified using digital elevation models in conjunction with imagery. They are also frequently identifiable through associations with each other (e.g., a dam and a reservoir).
 - b. Example Use Case: Identify all dams in a region.
 - c. Use Case Objectives:
 - i. Identify and extract vector representation (point, line, polygon) of these built water features.
 - ii. Where associated with existing features, put into correct spatial relationship (e.g., culverts and road, dams and waterbodies).
 - iii. Characterize the physical attributes of these features.

- 6) Aeronautical Infrastructure (Helipad, Runway, taxiway, apron)
 - a. Rationale (internal): FG collects aeronautical features to support Safety of Navigation and for display on topographic map products. These are typically disconnected asphalt or concrete areas that traditional LULC classifications identify simply as built areas, without further classification.
 - b. Example Use Case: Identify runways in association with the ground transportation network for mixed air-land transportation planning.
 - c. Use Case Objectives:
 - i. Identify and extract vector representations of on-ground aeronautical features.
 - ii. Characterize the physical attributes of these features.

THREE PRIMARY AFE FUNCTIONS

A) Object Detection (finding the object)

- a. This is the standard computer vision capability for identifying that something exists in an image, either providing a return as a point, bounding box, or other form that does not represent the real-world dimensions of the object.

B) Geometry Extraction (creating a vector representation of the object)

- a. This is typically what we mean by AFE and is a follow-on from A. This is the correct extraction of the feature from the imagery representing its real-world dimension in accordance with some extraction standards (e.g., the centerline of a road, the footprint of building, the roofline of a building).

C) Attribution (characterizing and providing information about the object)

- a. This is the addition of information to a geometry that reflects informative characteristics about the feature. It can overlap with A based on how fine-detailed the feature type is that is being detected (e.g., a detector for Buildings would need an attribute to determine if it is a hospital, but a detector for Hospitals would implicitly have that attribute). Typically, the attributes which are needed is driven by the data schema used for each mission set.
 - i. This could be further sub-divided into categorization (e.g., hospital, factory) and measurement (e.g., height, width).